United States Patent Application Publication	20250283905
Kind Code	A1
Publication Date	September 11, 2025
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TRANSPORT MECHANISM AND ANALYSIS APPARATUS

Abstract

A transport mechanism transports an analysis chip including a body part and a pair of engagement ribs in a transport direction orthogonal to the width direction, the transport mechanism including: a pair of screw shafts disposed parallel to each other along the transport direction, each of which has a shaft body and a helical protrusion helically formed on an outer peripheral surface of the shaft body, and propels the analysis chip in the transport direction by rotating about their axes while the helical protrusions are engaged with the engagement ribs; a transport table that extends along the transport direction and has a plurality of stop positions where the analysis chip is stopped; and a processor that controls rotation of the screw shafts to control movement of the analysis chip between the plurality of stop positions and stopping of the analysis chip at the plurality of stop positions.

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Family ID: **1000008629940**

Appl. No.: **19/218383**

Filed: **May 26, 2025**

Foreign Application Priority Data

JP	2022-192253	Nov. 30, 2022

Related U.S. Application Data

parent WO continuation PCT/JP2023/042081 20231122 PENDING child US 19218383

Publication Classification

Int. Cl.: **G01N35/04** (20060101)

U.S. Cl.:

CPC **G01N35/04** (20130101); G01N2035/0487 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/JP2023/042081, filed on Nov. 22, 2023, which claims priority from Japanese Patent Application No. 2022-192253, filed on Nov. 30, 2022. The entire disclosure of each of the above applications is incorporated herein by reference.

BACKGROUND
1. Technical Field
[0002] The present disclosure relates to a transport mechanism and an analysis apparatus.
2. Description of the Related Art
[0003] A micro-total analysis system (uTAS) is known as an analysis apparatus that analyzes a specimen such as a biological sample. The uTAS is an analysis apparatus that uses an analysis chip comprising a micro flow channel, a reaction chamber, a mixing chamber, and the like to perform a series of chemical or biochemical analysis steps within the analysis chip, such as mixing of a specimen and a reagent, a chemical reaction or a biochemical reaction, and detection of a product after the reaction. The analysis chip is loaded into the analysis apparatus, and various processes are executed within the analysis apparatus, such as a specimen dispensing process of dispensing a specimen, a reagent dispensing process of dispensing a reagent, a process for causing a specimen and a reagent to react with each other, and an optical detection process. The analysis chip is transported to each processing unit that executes various processes by a transport mechanism provided in the analysis apparatus.
[0004] As a transport mechanism that transports an object to be transported, a configuration is generally known in which the object to be transported is placed on a transport belt, the transport belt is rotated, and the object to be transported is transported for each transport belt, as in a belt conveyor. With the belt conveyor, the object to be transported is stationarily placed on the transport belt and does not move relative to the transport belt, so that the object to be transported can be stably transported. On the other hand, since the belt conveyor needs to be rotated, a structure such as a space for rotating the transport belt and a mechanism for driving the transport belt is required on a back surface of a transport passage, and space saving is difficult.
[0005] As a space-saving transport device, a transport device that comprises a screw shaft that includes a shaft body disposed along a transport direction and a ridge helically disposed on an outer periphery of the shaft body, and that transports an object to be transported by rotating the screw shaft is disclosed in JP2013-184801A.
[0006] For example, the transport device of JP2013-184801A holds a holder between a pair of screw shafts disposed in parallel, and sends the holder along a transport path to transport a specimen container held by the holder.
SUMMARY
[0007] The transport device of JP2013-184801A transports a holder that holds a specimen container accommodating a specimen in an upright state, and the specimen container is assumed to be a blood collection tube.
[0008] On the other hand, the analysis chip for uTAS is, for example, a microchannel device composed of a resin plate having a groove formed on one surface and a resin film bonded to cover

the groove, and has a shape significantly different from that of a cylindrical test container such as the blood collection tube assumed in JP2013-184801A. Therefore, it is difficult to directly apply the transport device of JP2013-184801A to transporting an analysis chip having a flat bottom surface, such as the analysis chip for uTAS.

[0009] In addition, in a case of transporting the analysis chip for uTAS, it is necessary to stop the analysis chip accurately at a stop position provided corresponding to each processing unit that is disposed along the transport passage and executes various processes.

[0010] An object of the present disclosure is to provide a transport mechanism and an analysis apparatus capable of stably transporting an analysis chip and accurately stopping the analysis chip at a stop position.

[0011] According to the present disclosure, there is provided a transport mechanism transports an analysis chip including a body part and a pair of engagement ribs provided to protrude outward from both ends in a width direction of the body part in a transport direction orthogonal to the width direction, the transport mechanism comprising: a pair of screw shafts disposed parallel to each other along the transport direction, each of which has a shaft body and a helical protrusion helically formed on an outer peripheral surface of the shaft body, and the pair of screw shafts propels the analysis chip in the transport direction by rotating about their respective axes while the helical protrusion are engaged with the engagement ribs; a transport table that extends along the transport direction and has a plurality of stop positions where the analysis chip is stopped; and a processor that controls rotation of the screw shafts to control movement of the analysis chip between the plurality of stop positions and stopping of the analysis chip at the plurality of stop positions.

[0012] The transport mechanism may further comprise: a positioning mechanism that positions the analysis chip, which is stopped at the stop position, at a preset position in the stop position, and it is preferable that the positioning mechanism includes a positioning pin that is inserted through a through-hole penetrating the body part of the analysis chip in an up-down direction, a positioning hole that is provided at the stop position of the transport table and into which a tip of the positioning pin inserted through the through-hole is inserted, and a lifting mechanism that lifts and lowers the positioning pin between an insertion position where the positioning pin is inserted through the through-hole and the tip is inserted into the positioning hole and a retreat position where the positioning pin is moved upward from the insertion position and is retreated from the insertion position.

[0013] The processor may be configured to rotate the screw shafts in a forward direction to transport the analysis chip to the stop position and stop the analysis chip, and then reversely rotate the screw shafts to return a phase position of the helical protrusion engaged with the analysis chip to an upstream side in the transport direction within a pitch range of the helical protrusion and release the engagement between the engagement ribs of the analysis chip and the helical protrusions.

[0014] In a case of rotating the screw shaft in a forward direction to transport the analysis chip to the stop position, the processor may be configured to stop the transport of the analysis chip at a position where the through-hole of the analysis chip and the positioning hole at least partially overlap each other and a center of the through-hole is located upstream of a center of the positioning hole in the transport direction, and cause the positioning mechanism to lower the positioning pin to the insertion position to send the analysis chip to a downstream side in the transport direction within a pitch range of the helical protrusion and position the analysis chip at the preset position.

[0015] It is preferable that a bottom surface of the analysis chip is flat, and the transport table has a transport surface on which the analysis chip is placed with the bottom surface in contact with the transport surface, and the analysis chip is transported while sliding on the transport surface.

[0016] It is preferable that the transport mechanism further comprises: a pressing part that presses the engagement rib of the analysis chip placed on the transport table from above.

[0017] It is preferable that the pressing part is provided at least at the stop position.
[0018] In a case in which the transport table has three or more stop positions as the plurality of stop positions, at least one of intervals between the stop positions adjacent to each other may be different from the other interval, and portions of the screw shaft in a length direction corresponding to the different intervals may have different pitches of the helical protrusions.
[0019] According to the present disclosure, there is provided an analysis apparatus comprising: the transport mechanism according to the present disclosure; and a plurality of processing units disposed to correspond to the plurality of stop positions and performing predetermined processing on a specimen accommodated in the analysis chip.
[0020] With the transport mechanism and the analysis apparatus of the technology of the present disclosure, it is possible to stably transport the analysis chip and accurately stop the analysis chip at the stop position.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0021] FIG. **1** is a schematic view showing a schematic configuration of an analysis apparatus.
[0022] FIG. **2** is a perspective view of an analysis chip with a bottom surface facing up.
[0023] FIG. **3** is an exploded perspective view of the analysis chip.
[0024] FIG. **4** is a perspective view of the analysis chip shown in FIG. **2** with its top and bottom reversed.
[0025] FIG. **5** is a perspective view showing a schematic configuration of a transport mechanism.
[0026] FIG. **6** is an enlarged plan view showing a part of the transport mechanism.
[0027] FIG. **7** is an enlarged perspective view showing a part of a transport table on which the analysis chip is placed.
[0028] FIG. **8** is a cross-sectional view showing the analysis chip that is stopped at a stop position and is positioned by a positioning pin.
[0029] FIG. **9** is a diagram showing a positional relationship between the analysis chip and a screw shaft immediately after the stop.
[0030] FIG. **10** is a diagram showing a positional relationship between the analysis chip and the screw shaft after the screw shaft is rotated reversely.
[0031] FIG. **11** is a diagram showing a state in which the analysis chip is stopped at a stop position in a state where a center of a through-hole of the analysis chip is slightly shifted to a downstream side with respect to a center of a positioning hole.
[0032] FIG. **12** is a diagram for describing a problem that occurs in a case in which positioning is performed by a positioning mechanism in the state of FIG. **11**.
[0033] FIG. **13**A is a diagram showing a state immediately after the analysis chip is stopped, and FIG. **13**B is a diagram showing a state after the positioning mechanism has performed positioning.
[0034] FIG. **14** is a perspective view showing a part of the transport mechanism comprising a pressing part.
[0035] FIG. **15** is a cross-sectional view taken along a line XV-XV of FIG. **14**.
[0036] FIG. **16** is an explanatory diagram of a transport mechanism of a modification example.
DESCRIPTION OF EMBODIMENTS
[0037] Hereinafter, a transport mechanism and an analysis apparatus comprising the transport mechanism according to one embodiment of the present disclosure will be described with reference to the drawings.
[0038] FIG. **1** is a schematic view showing a schematic configuration of an analysis apparatus **10** comprising a transport mechanism **14**. The analysis apparatus **10** is, for example, an immunological analysis apparatus that detects a test target substance in a specimen collected from a living body by

utilizing an antigen-antibody reaction. The analysis apparatus **10** dispenses a specimen and a reagent to an analysis chip **12**, causes mixing and an immune reaction in a microchannel in the analysis chip **12**, optically detects the test target substance in the specimen, and outputs a test result.

[0039] The analysis chip **12** is a microchannel device comprising a microchannel **32** therein (see FIG. **2**), and is, for example, an analysis chip for uTAS. As an example, a reagent such as a labeled antibody solution and a buffer solution is introduced into the microchannel **32** in addition to the specimen. In the microchannel **32**, an immune complex containing the test target substance in the specimen is generated and bound/free (B/F) separated by isotachophoresis, and separation and fractionation are performed by capillary electrophoresis. Then, the immune complex is irradiated with excitation light, and the fluorescence intensity corresponding to the concentration of the test target substance in the specimen is measured. The specimen is, for example, a biological sample such as blood collected from a living body. First, details of the analysis chip **12** will be described.

[Analysis Chip]

[0040] FIG. **2** is a perspective view of the analysis chip **12** with a bottom surface **12**Aa facing up. FIG. **3** is an exploded perspective view of the analysis chip **12**. FIG. **4** is a perspective view of the analysis chip **12** shown in FIG. **2** with its top and bottom reversed.

[0041] As shown in FIG. **2**, the analysis chip **12** comprises a body part **12**A and a pair of engagement ribs **12**B. In the body part **12**A, one surface **12**Aa is flat. For convenience of description, in FIG. **2**, the analysis chip **12** is shown with the bottom surface **12**Aa facing up. In a case in which the analysis chip **12** is transported by the transport mechanism **14**, the analysis chip **12** is transported with the one surface **12**Aa serving as the bottom surface (see FIG. **4**). Accordingly, this one surface **12**Aa is referred to as the bottom surface **12**Aa of the body part **12**A of the analysis chip **12**. That is, the analysis chip **12** comprises the body part **12**A of which the bottom surface **12**Aa is flat. The pair of engagement ribs **12**B are provided to protrude outward from both ends in a width direction of the body part **12**A. Here, the width direction of the body part **12**A is a direction orthogonal to a transport direction of the analysis chip **12**. In addition, the analysis chip **12** comprises a pair of support ribs **12**C that serve as a support portion in a case in which the analysis chip **12** is placed in a posture shown in FIG. **2**. The support ribs **12**C are provided to protrude outward from both ends in the width direction of the body part **12**A, as with the engagement rib **12**B, but a protruding width is smaller than that of the engagement rib **12**B. The analysis chip **12** comprises the microchannel **32** and a plurality of wells **33** for dispensing a specimen, a reagent, and the like in the microchannel **32**.

[0042] As shown in FIG. **3**, the analysis chip **12** is composed of a resin plate **30** and a film **31**.

[0043] The resin plate **30** is provided with a groove **34** constituting the microchannel **32** (see FIG. **2**) and the plurality of wells **33**. The microchannel **32** is, for example, a flow channel having a width of several tens of μm, and the groove **34** is a groove having a width of several tens of μm. The well **33** consists of a through-hole **33***a* that penetrates from one surface **30***a* of the resin plate **30** to the other surface **30***b* which is a back surface thereof, and a cylindrical portion **33***b* where an opening peripheral edge of the through-hole **33***a* protrudes from a surface on the other surface **30***b* side.

[0044] The film **31** is bonded to the one surface **30***a* of the resin plate **30** on which the groove **34** and the well **33** are formed so as to cover the groove **34** and the through-hole **33***a* (see FIGS. **2** and **3**). As a result, a portion of the groove **34** and the through-hole **33***a* that is open on the one surface **30***a* is sealed. The one surface **30***a* of the resin plate **30** to which the film **31** is bonded corresponds to the bottom surface **12**Aa of the analysis chip **12** (see FIG. **2**).

[0045] The pair of engagement ribs **12**B and support ribs **12**C described above are formed on the resin plate **30**. In addition, the resin plate **30** comprises a through-hole **36** that is separate from the through-hole **33***a* of the well **33** and that penetrates the resin plate **30** in an up-down direction in an end part region of the body part **12**A adjacent to the engagement rib **12**B. The through-hole **36** is a hole through which a positioning pin **62** of the transport mechanism **14**, which will be described below, is inserted. In this example, the through-holes **36** are provided at two positions. The through-hole **36** is provided at a position adjacent to the engagement rib **12**B in the end part region in the width direction of the body part **12**A.

[0046] As shown in FIG. **4**, the analysis chip **12** is transported in a transport direction A orthogonal to the width direction with the film **31** side as a bottom surface. In addition, the analysis chip **12** is transported in a position in which the engagement rib **12**B is at a front end located on a downstream side in the transport direction A, and the support rib **12**C is at a rear end located on an upstream side in the transport direction A. A planar shape of the analysis chip **12** is such that a front end portion **12**E located on the downstream side in the transport direction A is a straight line in the direction orthogonal to the transport direction A (see FIG. **6**).

[0047] Returning to FIG. **1**, a configuration of the analysis apparatus **10** will be described. The analysis apparatus **10** comprises, for example, a stocker **13** that accommodates a plurality of the analysis chips **12**, the transport mechanism **14** that transports the analysis chip **12**, a plurality of processing units **15** to **19** that perform predetermined processing on a specimen accommodated in the analysis chip **12**, a chip disposal unit **20**, and a processor **21**.

[0048] The transport mechanism **14** comprises a transport table **40** extending along the transport direction A of the analysis chip **12** and a pair of screw shafts **50** and **52** (see FIG. **5**). Although details of the transport mechanism **14** will be described below, the transport table **40** has a plurality of stop positions P**1** to P**5** where the analysis chip **12** is stopped. In addition, in this example, the transport mechanism **14** comprises a positioning mechanism **60** that positions the analysis chip **12**, which is stopped at the stop positions P**1** to P**5**, at preset positions in the stop positions P**1** to P**5**. The positioning mechanism **60** comprises a positioning pin **62** that positions the analysis chip **12** by inserting its tip into a positioning hole provided at each of the stop positions P**1** to P**5** of the transport table **40**, and a lifting mechanism **64** that lifts and lowers the positioning pin **62**. The lifting mechanism **64** lifts the positioning pin **62** between an insertion position where the positioning pin **62** is inserted through the through-hole **36** of the analysis chip **12** and the tip is inserted into the positioning hole and a retreat position where the positioning pin **62** is moved upward from the insertion position and is retreated from the insertion position. The lifting mechanism **64** is, for example, an eccentric cam.

[0049] The plurality of analysis chips **12** are stored in the stocker **13** in advance. The analysis chip **12** is taken out from the stocker **13** by a chipset mechanism (not shown) and is set on the transport table **40**. The stocker **13** is disposed below an upstream end of the transport table **40**, and the analysis chip **12** passes through a chipset opening **48** provided at the upstream end of the transport table **40** and is set on a transport surface **42** (see FIG. **5**).

[0050] The plurality of processing units **15** to **19** are, for example, a specimen dispensing processing unit **15**, a reagent dispensing processing unit **16**, a pressurization processing unit **17**, a reaction processing unit **18**, and a detection processing unit **19**. The plurality of processing units **15** to **19** are disposed to correspond to the respective stop positions P**1** to P**5** along the transport direction A of the analysis chip **12**.

[0051] The specimen dispensing processing unit **15** comprises a dispensing mechanism **15***b* comprising a nozzle **15***a*, and dispenses a specimen into the microchannel **32** of the analysis chip **12** stopped at the stop position P**1**. A specimen collection tube (not shown) accommodating the collected specimen is loaded into the analysis apparatus **10**, and the nozzle **15***a* sucks the specimen from the specimen collection tube and dispenses the specimen into the microchannel **32** from the predetermined well **33** of the analysis chip **12**.

[0052] The reagent dispensing processing unit **16** comprises a dispensing mechanism **16***b* comprising a nozzle **16***a*, and dispenses a reagent into the microchannel **32** of the analysis chip **12** stopped at the stop position P**2**. A reagent container (not shown) accommodating the reagent is loaded into the analysis apparatus **10**, and the nozzle **16***a* sucks the reagent from the reagent container and dispenses the reagent into the microchannel **32** from the predetermined well **33** of the analysis chip **12**.

[0053] In the pressurization processing unit **17**, gas is fed through at least a part of the plurality of wells **33** from the outside of the analysis chip **12** to introduce the specimen and the reagent into the microchannel **32** and mix them, so that the inside of the microchannel **32** is pressurized.

[0054] In the reaction processing unit **18**, a labeled antibody is concentrated, an antigen-antibody reaction is performed, and B/F separation is performed by isotachophoresis.

[0055] In the detection processing unit **19**, the separation and fractionation of the immune complex are performed by capillary gel electrophoresis method, and then detection processing of detecting the test target substance in the specimen is executed. For example, the test target substance contained in the immune complex is detected by a laser fluorescence method. The detection processing unit **19** comprises an optical head **19***a*. The optical head **19***a* is disposed at a position where fluorescence from a predetermined location of the microchannel **32** can be received while irradiating the predetermined location with laser light from the bottom surface **12**Aa of the analysis chip **12** in a state where the analysis chip **12** is stopped at the stop position P**5**. The optical head **19***a* comprises, for example, an irradiation section and a light receiving section.

[0056] The analysis apparatus **10** further comprises a pneumatic section (not shown) that pressurizes and depressurizes the inside of the flow channel and an electrophoresis section **22** for electrophoresis. During the pressurization processing, the reaction processing, and the detection processing, a pressing unit **24** comprising a sealing portion and a supply pipe presses a surface of the analysis chip **12**, which comprises the well **33**, in order to seal the well **33** or to connect a gas supply pipe to the well **33**. The pressing unit **24** can be lifted and lowered by the lifting mechanism between a position where it is separated from the analysis chip **12** and a position where it presses the analysis chip **12**. In this example, the lifting mechanism **64** provided in the positioning mechanism **60** of the transport mechanism **14** also serves as the lifting mechanism of the pressing unit **24**.

[0057] The chip disposal unit **20** accommodates the analysis chip **12** after the analysis has been completed. The chip disposal unit **20** is disposed at a location where the analysis chip **12** that falls from a disposal opening **49** provided at a downstream end of the transport table **40** is received.

[0058] The processor **21** constitutes a control unit that integrally controls each unit of the analysis apparatus **10**. An example of the processor **21** is a central processing unit (CPU) which performs various types of control by executing a program. The CPU functions as a control unit which controls each unit by executing a program. The processor **21** also serves as a control unit of the transport mechanism **14**. The processor **21** functions as a control unit of the transport mechanism **14**, and controls a driving unit **54** of the transport mechanism **14**, which will be described below. As a result, the processor controls rotation of the screw shafts **50** and **52** to control movement of the analysis chip **12** between the plurality of stop positions P**1** to P**5** and stopping of the analysis chip **12** at the plurality of stop positions P**1** to P**5**.

[0059] Hereinafter, the transport mechanism **14** will be described in detail.

[Transport Mechanism]

[0060] FIG. **5** is a perspective view showing a schematic configuration of the transport mechanism **14**, and FIG. **6** is an enlarged plan view showing a part of the transport mechanism **14**. In addition, FIG. **7** is an enlarged perspective view showing a part of the transport table **40** on which the analysis chip **12** is placed. The transport mechanism **14** transports the above-described analysis chip **12** in the transport direction A orthogonal to the width direction of the analysis chip **12**. The transport mechanism **14** comprises the transport table **40** and the pair of screw shafts **50** and **52**.

[0061] The transport table **40** has the transport surface **42** on which the bottom surface **12**Aa of the analysis chip **12** is placed with the bottom surface **12**Aa in contact with the transport surface **42**. The analysis chip **12** is transported while sliding on the transport surface **42**. That is, the analysis chip **12** is transported by sliding on the transport surface **42** of the transport table **40** in a state where the bottom surface **12**Aa of the analysis chip **12** is in contact with the transport surface **42**. A temperature of the transport surface **42** may be controlled by temperature control means (not shown). The temperature control means comprises, for example, a heater and a temperature control circuit. In a case in which the analysis chip **12** is transported with the bottom surface **12**Aa in contact with the temperature-controlled transport surface **42**, the specimen and the reagent accommodated in the analysis chip **12** can be transported while being temperature-controlled. A region of the transport surface **42** with which the bottom surface **12**Aa of the analysis chip **12** comes into contact in a case of transporting the analysis chip **12** is referred to as a contact region **42**A. The contact region **42**A is a region between the pair of screw shafts **50** and **52**, and is a region that extends along the transport direction A with a width equivalent to a length L in the width direction of the body part **12**A of the analysis chip **12**. In FIGS. **5** to **7** and the like, a region between two-dot chain lines **41** on the transport surface **42** corresponds to the contact region **42**A.

[0062] The transport surface **42** has a hole **43** formed in the contact region **42**A and a recessed portion **45** formed in at least a part of a periphery of an opening edge **44** of the hole **43**. The hole **43** formed in the contact region **42**A of the transport surface **42** is a positioning hole into which the tip of the positioning pin **62** is inserted, a hole for fixing screw insertion for fixing the transport table **40** in the analysis apparatus **10**, or the like.

[0063] The recessed portion **45** has a recessed surface **46** that makes a height of a partial opening edge **44***a*, which is located on the downstream side in the transport direction A on the entire circumference of the opening edge **44** and includes a contact point C with a tangent B extending in the direction orthogonal to the transport direction A, lower than the transport surface **42**, and the recessed surface **46** extends from the partial opening edge **44***a* to an outside of the contact region **42**A (see FIGS. **6** and **7**).

[0064] As shown in FIG. **5**, the pair of screw shafts **50** and **52** are disposed on the transport table **40**. The pair of screw shafts **50** and **52** are disposed parallel to each other along the transport direction A. The screw shaft **50** has a shaft body **50**A and a helical protrusion **50**B helically formed on an outer peripheral surface of the shaft body **50**A. Similarly, the screw shaft **52** has a shaft body **52**A and a helical protrusion **52**B helically formed on an outer peripheral surface of the shaft body **52**A.

[0065] A helical pitch SP of the helical protrusion **50**B of one screw shaft **50** and a helical pitch SP of the helical protrusion **52**B of the other screw shaft **52** are the same (see FIG. **6**). A direction of helical rotation of the helical protrusion **50**B of one screw shaft **50** is opposite to a direction of helical rotation of the helical protrusion **52**B of the other screw shaft **52**. One end of each of the pair of screw shafts **50** and **52** is connected to the driving unit **54** that rotationally drives the screw shafts **50** and **52**. The pair of screw shafts **50** and **52** are attached to the driving unit **54** such that phase positions of the helical protrusions **50**B and **52**B match each other. The driving unit **54** includes, for example, a motor **55** and a plurality of gears (not shown). One end of one screw shaft **50** is connected to the motor **55** such that a driving force of the motor **55** can be directly transmitted as a rotational force. One end of the other screw shaft **52** is connected to one of the plurality of gears, and the driving force of the motor **55** is transmitted as a rotational force via the plurality of gears. The plurality of gears are combined such that a rotation direction of one screw shaft **50** and a rotation direction of the other screw shaft **52** are opposite to each other. The other end of each of the screw shafts **50** and **52** is supported by a bearing unit **56**.

[0066] The pair of screw shafts **50** and **52** rotate about their axes while the respective helical protrusions **50**B and **52**B and the pair of engagement ribs **12**B are engaged with each other, thereby propelling the analysis chip **12** in the transport direction A.

[0067] As shown in FIGS. **5** and **6**, the analysis chip **12** is placed on the transport surface **42** of the transport table **40** in a posture in which the flat bottom surface **12**Aa (see FIG. **2**) is in contact with the transport surface **42** and the front end portion **12**E extending in the width direction is orthogonal to the transport direction A. In addition, the analysis chip **12** is placed between the pair of screw shafts **50** and **52** disposed parallel to each other. The helical protrusion **50**B is a series of convex portions helically formed along an axial direction, and a protrusion portion **50**Ba is defined as one convex portion in a case in which the helical protrusion **50**B is viewed in a plan view. A plurality of the protrusion portions **50**Ba are arranged at equal pitches SP along the axial direction in a case in which the helical protrusion **50**B is viewed in a plan view. Similarly, a protrusion portion **52**Ba is defined as one convex portion in a case in which the helical protrusion **52**B is viewed in a plan view. A plurality of the protrusion portions **52**Ba are arranged at equal pitches SP along the axial direction in a case in which the helical protrusion **52**B is viewed in a plan view. In the analysis chip **12**, one engagement rib **12**B is located in a recess portion between two protrusion portions **50**Ba adjacent to each other in the axial direction in a case in which a helical groove of the screw shaft **50**, more specifically, the helical protrusion **50**B is viewed in a plan view. Further, the other engagement rib **12**B is placed to be located in a recess portion between two protrusion portions **52**Ba adjacent to each other in the axial direction in a case in which a helical groove of the screw shaft **52**, more specifically, the helical protrusion **52**B is viewed in a plan view (see FIG. **6**). An interval of the recess portion between the two adjacent protrusion portions **50**Ba is narrower than the pitch SP of the helical protrusion **50**B by a width of one protrusion portion **50**Ba, but is substantially equal to the pitch SP. An interval between the two adjacent protrusion portions **52**Ba is also narrower than the pitch SP of the helical protrusion **52**B by a width of one protrusion portion **52**Ba, but is substantially equal to the pitch SP.

[0068] The engagement between the screw shafts **50** and **52** and the pair of engagement ribs **12**B will be described in more detail below. The engagement between the screw shaft **50** and one engagement rib **12**B will be described as an example. In a case in which the helical protrusion **50**B rotates due to the rotation of the screw shaft **50**, a phase of the protrusion portion **50**Ba, which is a convex portion of the helical protrusion **50**B, and a phase of the recess portion, which is a groove of the adjacent protrusion portion **50**Ba, move in the transport direction. As a result, the protrusion portion **50**Ba presses the engagement rib **12**B from the rear (upstream side), and propels the engagement rib **12**B in the axial direction of the screw shaft **50** in accordance with the change in phase of the protrusion portion **50**Ba while maintaining a contact state with the engagement rib **12**B. The same applies to the engagement between the screw shaft **52** and the other engagement rib **12**B. In this way, the screw shafts **50** and **52** are rotated about their axes while the helical protrusions **50**B and **52**B and the engagement ribs **12**B and **12**B are engaged with each other, whereby the engagement ribs **12**B and **12**B are pressed in the axial direction by the helical protrusions **50**B and **52**B. As a result, the analysis chip **12** is transported in the transport direction A which is the axial direction. In this case, the analysis chip **12** is transported while sliding on the transport surface **42** in a state where the bottom surface **12**Aa is in contact with the transport surface **42**.

[0069] As described above, the processor **21** controls the driving unit **54** to control the rotation of the screw shafts **50** and **52**, thereby controlling the movement of the analysis chip **12** between the plurality of stop positions P**1** to P**5** and the stopping of the analysis chip **12** at the plurality of stop positions P**1** to P**5**.

[0070] The analysis chip **12** is transported to each of the stop positions P**1** to P**5** (see FIG. **1**), and the above-described various processes are performed on the analysis chip **12** at each of the stop positions P**1** to P**5**. After the analysis chip **12** is transported to each of the stop positions P**1** to P**5**, the positioning mechanism **60** performs positioning before various processes are performed. FIG. **8** is a cross-sectional view showing the analysis chip **12** that is stopped at the stop position P**1** and is positioned by the positioning pin **62**. The positioning pin **62** is provided to face a positioning hole **43** among the holes **43** provided on the transport surface **42** of the transport table **40**. Here, the reference for the stop positions P**1** to P**5** is the positioning hole **43**. The positioning pin **62** is positioned at a retreat position above the transport table **40** during the transport of the analysis chip **12**, and is lowered by the lifting mechanism **64** during the positioning. The positioning pin **62** lowered by the lifting mechanism **64** is inserted through the through-hole **36** of the analysis chip **12**, and the tip thereof is inserted into the positioning hole **43**, as shown in FIG. **8**. In a case in which the positioning pin **62** is inserted through the through-hole **36**, a position of the analysis chip **12** in the transport direction is adjusted, so that the analysis chip **12** is positioned at a preset position in the stop position P**1**. Even at the stop positions P**2** to P**5** other than the stop position P**1**, the positioning hole **43** is provided on the transport surface **42** of the transport table **40**, and the positioning pin **62** of which the tip is inserted into the positioning hole **43** is provided.

[0071] Hereinafter, the expression "transporting the analysis chip **12** to each of the stop positions P**1** to P**5** and stopping the analysis chip **12**" means that the analysis chip **12** is stopped at a position where the through-hole **36** and the positioning hole **43** serving as a reference for the stop positions P**1** to P**5** face each other. Taking the stop position P**1** as an example, the stop position P**1** has a width in a range where the positioning hole **43** and the through-hole **36** at least partially overlap each other. As described above, the through-hole **36** of the analysis chip **12**, and the positioning pin **62** and the positioning hole **43** are provided to position the analysis chip **12** at a preset position in the stop position P**1**. As shown in FIG. **8**, the preset position is a position where the positioning pin **62** is inserted through the through-hole **36** of the analysis chip **12**, and is more specifically a position where the center of the through-hole **36** matches the center of the positioning pin **62** and the center of the positioning hole **43**. As shown in FIG. **8**, at each of the stop positions P**1** to P**5**, in a case in which the positioning pin **62** can be inserted into the through-hole **36**, the analysis chip **12** is positioned at a preset position in each of the stop positions P**1** to P**5**. In order to perform such positioning, it is necessary to have a clearance for adjusting a relative positional relationship in the transport direction between the analysis chip **12** and the helical protrusions **50**B and **52**B in a state where the analysis chip **12** is stopped at each of the stop positions P**1** to P**5**.

[0072] Therefore, the processor **21** executes the following control. The stop position P**1** will be described as an example. First, the processor **21** rotates the screw shafts **50** and **52** in a forward direction to transport the analysis chip **12** to the stop position P**1** and stop the analysis chip **12**. Here, the forward direction is a rotation direction in which the analysis chip **12** can be transported in the transport direction A. As described above, the analysis chip **12** is propelled in the transport direction A such that the rear ends of the engagement ribs **12**B are pressed by the helical protrusions **50**B and **52**B of the screw shafts **50** and **52**. Therefore, immediately after the rotation of the screw shafts **50** and **52** is stopped and the transport of the analysis chip **12** is stopped, a state where the rear ends of the engagement ribs **12**B are in contact with the helical protrusions **50**B and **52**B is maintained. This state is shown in detail in a circular enlarged view IX of FIG. **9**. Although the enlarged view IX shows the screw shaft **50** as an example, the same applies to the screw shaft **52**. The enlarged view IX shows a state where the helical protrusion **50**B of the screw shaft **50** and the rear end of the engagement rib **12**B of the analysis chip **12** are in contact with each other during the transport and the stop. Here, a state where the protrusion portion **50**Ba of the helical protrusion **50**B is in contact with the rear end of the engagement rib **12**B is referred to as a state where the helical protrusion **50**B and the engagement rib **12**B are engaged with each other.

[0073] In the present embodiment, the processor **21** transports the analysis chip **12** to the stop position P**1** and stops the analysis chip **12**, and then reversely rotates the screw shafts **50** and **52**. As shown in FIG. **10**, due to the reverse rotation of the screw shafts **50** and **52**, the phase positions of the helical protrusions **50**B and **52**B engaged with the analysis chip **12** are returned to the upstream side in the transport direction A within the range of the pitches SP of the helical protrusions **50**B and **52**B, and a state where the pair of engagement ribs **12**B of the analysis chip **12** and the helical protrusions **50**B and **52**B are engaged with each other is released. As a result, in the stop position P**1**, it is possible to ensure a clearance for adjusting the relative positional relationship in the transport direction between the helical protrusions **50**B and **52**B and the analysis chip **12**.

[0074] In a circular enlarged view X in FIG. **10**, a state of the helical protrusion **50**B immediately after the stop is shown by a broken line. The amount of rotation in a case in which the screw shafts

50 and **52** are reversely rotated is adjusted such that the amount of movement of the phase positions of the helical protrusions **50**B and **52**B is within the range of the pitch SP and both the rear end and the front end of the engagement rib **12**B do not come into contact with the helical protrusions **50**B and **52**B. The amount of rotation is, for example, approximately ½ rotation or less. In the example of FIG. **10**, the phase position of the helical protrusion **50**B is retreated by R in the axial direction. The distance R is about ½ or less of the pitch SP.
[0075] After a state where the engagement ribs **12**B of the analysis chip **12** and the helical protrusions **50**B and **52**B are engaged with each other is released, the positioning pin **62** is lowered to perform positioning (see FIG. **10**). At the stop position P**1**, the positioning pin **62** and the positioning hole **43**, and the through-hole **36** at least partially overlap each other. Even in a state where the centers of the positioning pin **62** and the positioning hole **43** do not match the center of the through-hole **36**, in a case in which the positioning pin **62** is inserted through the through-hole **36**, the analysis chip **12** is moved along the transport direction, and the position of the analysis chip **12** is adjusted. As a result, the analysis chip **12** is positioned at a position where the centers of the positioning pin **62** and the positioning hole **43** match the center of the through-hole **36**.
[0076] In a state where the analysis chip **12** is positioned at preset positions in the stop positions P**1** to P**5** of the transport table **40**, various processes are executed, such as dispensing a specimen at the stop position P**1** and dispensing a reagent at the stop position P**2**.
[0077] As described above, the present transport mechanism **14** comprises the pair of screw shafts **50** and **52** that propel the analysis chip **12** in the transport direction A by rotating about their axes while the helical protrusions **50**B and **52**B and the engagement ribs **12**B and **12**B are engaged with each other. With the present configuration, the analysis chip **12** can be stably transported. Since the screw shafts **50** and **52** are rotated to transport the analysis chip **12**, space saving can be achieved compared to a transport mechanism that requires a driving unit on a back surface side of a transport table as in a belt conveyor. In addition, since the processor **21** controls the rotation of the screw shafts **50** and **52** to control the movement of the analysis chip **12** between the plurality of stop positions P**1** to P**5** and the stopping of the analysis chip **12** at the plurality of stop positions P**1** to P**5**, the analysis chip **12** can be accurately stopped at the preset positions in the stop positions P**1** to P**5**.
[0078] As described above, in the present analysis apparatus **10**, the transport mechanism **14** comprises the positioning mechanism **60**. Since the positioning mechanism **60** inserts the positioning pin **62** through the through-hole **36** of the analysis chip **12** and the tip of the positioning pin **62** into the positioning hole **43**, the analysis chip **12** can be positioned at the stop positions P**1** to P**5** with higher accuracy. For example, even in a case in which the analysis chip **12** is stopped with the center of the through-hole **36** slightly deviated from the center of the positioning hole **43**, the analysis chip **12** is guided and positioned by the positioning pin **62** such that the center of the through-hole **36** matches the center position of the positioning pin **62** by inserting the positioning pin **62** into the through-hole **36**. As described above, even in a case in which the analysis chip **12** is slightly deviated and stopped, the analysis chip **12** is slightly moved and positioned such that the center of the through-hole **36** and the center of the positioning hole **43** match each other, and the analysis chip **12** can be positioned at the stop positions P**1** to P**5** with extremely high accuracy.
[0079] In addition, in the present embodiment, the processor **21** rotates the screw shafts **50** and **52** in the forward direction to transport the analysis chip **12** to the stop positions P**1** to P**5** and stop the analysis chip **12**, and then reversely rotates the screw shafts **50** and **52** before the positioning via the positioning mechanism **60**. Due to the reverse rotation of the screw shafts **50** and **52**, the phases of the helical protrusions **50**B and **52**B engaged with the analysis chip **12** are returned to the upstream side in the transport direction A, and a state where the engagement ribs **12**B of the analysis chip **12** and the helical protrusions **50**B and **52**B are engaged with each other is released.
[0080] For example, as shown in FIG. **11**, in a case in which the center of the through-hole **36** of the analysis chip **12** that is transported to and stopped at the stop position P**1** is slightly shifted to

the downstream side with respect to the center of the positioning hole **43**, in a case in which the positioning pin **62** is inserted through the through-hole **36** of the analysis chip **12** in this state, the analysis chip **12** is guided by the positioning pin **62** such that the center of the through-hole **36** and the center of the positioning hole **43** match each other, and a force in a direction of moving the analysis chip **12** to the upstream side is applied to the analysis chip **12**. In a state immediately after the rotation of the screw shafts **50** and **52** in the forward direction is stopped, as shown in FIG. **12**, the rear end of the engagement rib **12**B of the analysis chip **12** is in contact with the protrusion portion **50**Ba of the helical protrusion **50**B. Therefore, the movement of the analysis chip **12** to the upstream side is in a restricted state. In a case in which the positioning pin **62** is inserted into the through-hole **36** and a force is applied to shift the analysis chip **12** to the upstream side in a state where the rear end (upstream side) of the engagement rib **12**B is in contact with the protrusion portion **50**Ba of the helical protrusion **50**B and cannot move, a load is applied to the engagement rib **12**B, which may cause the engagement rib **12**B to be damaged. FIGS. **11** and **12** show a relationship between the screw shaft **50** and one engagement rib **12**B as an example, and the same applies to a relationship between the screw shaft **52** and the other engagement rib **12**B.

[0081] On the other hand, in the present embodiment, since the processor **21** performs control of releasing a state where the engagement ribs **12**B of the analysis chip **12** and the helical protrusions **50**B and **52**B are engaged with each other before the positioning via the positioning mechanism **60**, an over-constrained state of the analysis chip **12** in the positioning via the positioning pin **62** can be released, and the damage to the analysis chip **12** can be suppressed. Therefore, the occurrence of an error due to the damage to the analysis chip **12** during the analysis can be suppressed, and the analysis efficiency of the analysis apparatus **10** can be improved.

[0082] In the above description, an aspect in which the processor **21** rotates the screw shafts **50** and **52** in the forward direction to transport the analysis chip **12** to the stop positions P**1** to P**5** and stop the analysis chip **12**, and then reversely rotates the screw shafts **50** and **52** before the positioning via the positioning mechanism **60** has been described, but, instead of this aspect, the processor **21** may perform the following control.

[0083] Even in a modification example of the control of the processor **21**, the processor **21** rotates the screw shafts **50** and **52** in the forward direction to transport the analysis chip **12** to the stop positions P**1** to P**5** and stop the analysis chip **12**. In the modification example, the processor **21** performs control of stopping the analysis chip **12** at a position where the through-hole **36** of the analysis chip **12** and the positioning hole **43** of the transport table **40** at least partially overlap each other and the center of the through-hole **36** is on the upstream side in the transport direction A with respect to the center of the positioning hole **43** in the stop positions P**1** to P**5**. FIG. **13**A is a cross-sectional view showing a positional relationship between the analysis chip **12** and the positioning mechanism **60** in a state where the analysis chip **12** is stopped on the upstream side of the stop position P**1** in the transport direction A. As shown in FIG. **13**A, the analysis chip **12** is stopped at the stop position P**1** such that a center **36***a* of the through-hole **36** is on the upstream side in the transport direction A with respect to the stop position P**1**, which is a preset position and is the center of the positioning hole **43**. In FIG. **13**A, a distance between the center **36***a* of the through-hole **36** and the center of the positioning hole **43** is S.

[0084] In this state, in a case in which the positioning mechanism **60** lowers the positioning pin **62** to the insertion position, as shown in FIG. **13**B, the analysis chip **12** is guided by the positioning pin **62**, is sent to the downstream side in the transport direction A by the distance S, and is positioned at the preset position.

[0085] As described with reference to FIGS. **11** and **12**, in a case in which the positioning mechanism **60** performs the positioning in a state where the analysis chip **12** is stopped with the center of the through-hole **36** shifted to the downstream side with respect to the center of the positioning hole **43**, the analysis chip **12** may be damaged. However, as described above, by controlling the analysis chip **12** such that the analysis chip **12** is stopped at a position where the center of the through-hole **36** is on the upstream side in the transport direction A with respect to the center of the positioning hole **43**, the analysis chip **12** is moved to the downstream side during the positioning via the positioning mechanism **60**, so that no load is applied to the engagement rib **12**B. Accordingly, the damage to the analysis chip **12** can be suppressed.

[0086] In addition, as shown in FIG. **14**, the transport mechanism **14** may further comprise a pressing part **70** that presses the engagement rib **12**B of the analysis chip **12** placed on the transport table **40** from above. FIG. **15** is a cross-sectional view taken along a line XV-XV in FIG. **14**.

[0087] As shown in FIGS. **14** and **15**, the pressing part **70** is formed by, for example, bending a single plate-shaped member. The pressing part **70** presses the engagement rib **12**B and the support rib **12**C of the analysis chip **12** from above at one end part **72**, is disposed to straddle the screw shafts **50** and **52**, and is fixed to the transport table **40** at the other end part **74**. A screw hole is provided in the other end part **74**, and the other end part **74** is fixed to the transport table **40** by screwing with a screw **76**.

[0088] As described above, during the pressurization processing, the reaction processing, and the detection processing, the pressing unit **24** comprising a sealing portion and a supply pipe presses the surface of the analysis chip **12**, which comprises the well **33**, in order to seal the well **33** or to connect a gas supply pipe to the well **33**. After performing each processing at each of the stop positions P**1** to P**5**, the pressing unit **24** is lifted to release the contact state between the pressing unit **24** and the analysis chip **12** before transporting the analysis chip **12** to the next stop positions P**1** to P**5**. In this case, it is considered that the analysis chip **12** adheres to the pressing unit **24** and floats up from the transport table **40**, and then is peeled off from the pressing unit **24** and falls. In this way, in a case in which the analysis chip **12** floats up and then falls, the impact of the falling may cause liquid splashing, and contamination may occur, which may induce a measurement error. By providing the pressing part **70**, it is possible to prevent the analysis chip **12** from floating up in this manner, and as a result, it is possible to suppress a decrease in the analysis accuracy of the analysis apparatus **10**.

[0089] It is preferable that the pressing part **70** is provided at each of the stop positions P**1** to P**5**. The pressing part **70** may be provided at positions other than the stop positions P**1** to P**5**.

[0090] In the transport mechanism **14** described above, intervals D between stop positions adjacent to each other among the plurality of stop positions P**1** to P**5** are all the same and are equal intervals. Therefore, the pitches SP of the helical protrusions **50**B and **52**B of the screw shafts **50** and **52** are constant. On the other hand, as in a transport mechanism **114** of a modification example shown in FIG. **16** compared to the transport mechanism **14**, the intervals between the stop positions P**1** to P**5** may be different. In FIG. **16**, only one screw shaft **50** of the pair of screw shafts **50** and **52** of the transport mechanism **14** is shown, and, similarly, only one screw shaft **150** of a pair of screw shafts of the transport mechanism **114** is shown. Similarly with the screw shaft **50**, the screw shaft **150** has a shaft body **150**A and a helical protrusion **150**B helically formed on an outer peripheral surface of the shaft body **150**A. In the transport mechanism **114**, an interval D**1** between the stop position P**1** and the stop position P**2** is the same as an interval D**1** between the stop position P**2** and the stop position P**3**, but is different from an interval D**2** between the stop position P**3** and the stop position P**4**. For example, in a case in which the spaces of the specimen dispensing processing unit **15** and the reagent dispensing processing unit **16** can be narrower than those of the other processing units **17** to **19**, as in the transport mechanism **114** of FIG. **16**, the interval D**1** between the stop position P**1** and the stop position P**2** and the interval D**1** between the stop position P**2** and the stop position P**3** may be shorter than the interval D**2** between the stop position P**3** and the stop position P**4** and an interval D**2** between the stop position P**4** and the stop position P**5**.

[0091] In the transport mechanism **114**, a pitch SP**1** of the helical protrusion **150**B in a portion corresponding to the interval D**1** of the screw shaft **150** is different from a pitch SP**2** in a portion corresponding to the interval D**2**. The pitch SP**1** is an interval between protrusion portions **150**Ba adjacent to each other in the interval D**1**, and the pitch SP**2** is an interval between protrusion

portions **150**Ba adjacent to each other in the interval D**2**. Here, the term "a portion" means a portion of the screw shaft **150**, means a part over the entire region of the interval D**1** (or the interval D**2**) of the screw shaft **150**, and does not mean a part in the interval D**1**. The pitch SP**1** in the relatively short interval D**1** is smaller than the pitch SP**2** in the relatively long interval D**2**.
[0092] In this way, in a case in which three or more stop positions P**1** to P**5** are provided, at least one of intervals between the stop positions adjacent to each other may be different from the other interval, and portions of the screw shaft **150** in a length direction corresponding to the different intervals may have different pitches of the helical protrusions **150**B. The intervals between the stop positions P**1** to P**5** can be set according to the processing contents of each processing unit, and space saving can be achieved. As shown in FIG. **16**, the transport mechanism **114** in which the intervals between the stop positions P**1** to P**5** are set to be irregular intervals according to the processing contents can shorten the transport passage compared to the transport mechanism **14** in which the intervals between the stop positions P**1** to P**5** are set to be regular intervals.
[0093] The transport mechanism **14** can be applied to an analysis apparatus other than the analysis apparatus **10** described above, and the analysis chip that can be transported by the transport mechanism **14** is not limited to the analysis chip **12**. The analysis chip transported by the transport mechanism **14** may be any analysis chip as long as it comprises at least a body part having a flat bottom surface and a pair of engagement ribs provided to protrude outward from both ends in a width direction of the body part.
[0094] In addition, in the above-described embodiment, as a hardware structure of the processor, various processors shown below can be used. The various processors include, in addition to a CPU that is a general-purpose processor that executes software (program) to function as various processing units, a programmable logic device (PLD) of which a circuit configuration can be changed after manufacturing, such as a field-programmable gate array (FPGA), and a dedicated electric circuit that is a processor having a circuit configuration dedicatedly designed for executing specific processing, such as an application specific integrated circuit (ASIC).
[0095] The above-described processing may be executed by one of these various processors, or a combination of two or more processors of the same type or different types (for example, a combination of a plurality of FPGAs and a combination of a CPU and an FPGA). In addition, a plurality of processing units may be configured of one processor. As an example in which the plurality of processing units are configured of one processor, there is a form in which a processor that realizes all functions of a system including the plurality of processing units by using one integrated circuit (IC) chip is used, such as a system on chip (SOC).
[0096] Furthermore, the hardware structure of the processors is, more specifically, an electric circuit (circuitry) in which circuit elements such as semiconductor elements are combined.
[0097] The following appendices are further disclosed with respect to the above embodiment.
APPENDIX 1
[0098] A transport mechanism that transports an analysis chip including a body part and a pair of engagement ribs provided to protrude outward from both ends in a width direction of the body part in a transport direction orthogonal to the width direction, the transport mechanism comprising:
[0099] a pair of screw shafts disposed parallel to each other along the transport direction, each of which has a shaft body and a helical protrusion helically formed on an outer peripheral surface of the shaft body, and the pair of screw shafts propels the analysis chip in the transport direction by rotating about their respective axes while the helical protrusions are engaged with the engagement ribs; [0100] a transport table that extends along the transport direction and has a plurality of stop positions where the analysis chip is stopped; and [0101] a processor that controls rotation of the screw shafts to control movement of the analysis chip between the plurality of stop positions and stopping of the analysis chip at the plurality of stop positions.
APPENDIX 2
[0102] The transport mechanism according to Appendix 1, further comprising: [0103] a positioning

mechanism that positions the analysis chip, which is stopped at the stop position, at a preset position in the stop position, [0104] in which the positioning mechanism includes a positioning pin that is inserted through a through-hole penetrating the body part of the analysis chip in an up-down direction, a positioning hole that is provided at the stop position of the transport table and into which a tip of the positioning pin inserted through the through-hole is inserted, and a lifting mechanism that lifts and lowers the positioning pin between an insertion position where the positioning pin is inserted through the through-hole and the tip is inserted into the positioning hole and a retreat position where the positioning pin is moved upward from the insertion position and is retreated from the insertion position.

APPENDIX 3

[0105] The transport mechanism according to Appendix 2, [0106] in which the processor is configured to rotate the screw shafts in a forward direction to transport the analysis chip to the stop position and stop the analysis chip, and then reversely rotate the screw shafts to return a phase position of the helical protrusion engaged with the analysis chip to an upstream side in the transport direction within a pitch range of the helical protrusion and release the engagement between the engagement ribs of the analysis chip and the helical protrusions.

APPENDIX 4

[0107] The transport mechanism according to Appendix 2, [0108] in which, in a case of rotating the screw shaft in a forward direction to transport the analysis chip to the stop position, the processor is configured to [0109] stop the transport of the analysis chip at a position where the through-hole of the analysis chip and the positioning hole at least partially overlap each other and a center of the through-hole is located upstream of a center of the positioning hole in the transport direction, and [0110] cause the positioning mechanism to lower the positioning pin to the insertion position to send the analysis chip to a downstream side in the transport direction within a pitch range of the helical protrusion and position the analysis chip at the preset position.

APPENDIX 5

[0111] The transport mechanism according to any one of Appendices 1 to 4, [0112] in which a bottom surface of the analysis chip is flat, and [0113] the transport table has a transport surface on which the analysis chip is placed with the bottom surface in contact with the transport surface, and the analysis chip is transported while sliding on the transport surface.

APPENDIX 6

[0114] The transport mechanism according to any one of Appendices 1 to 5, further comprising: [0115] a pressing part that presses the engagement rib of the analysis chip placed on the transport table from above.

APPENDIX 7

[0116] The transport mechanism according to Appendix 6, [0117] in which the pressing part is provided at least at the stop position.

APPENDIX 8

[0118] The transport mechanism according to any one of Appendices 1 to 7, [0119] in which the transport table has three or more stop positions as the plurality of stop positions, and [0120] at least one of intervals between the stop positions adjacent to each other is different from the other interval, and portions of the screw shaft in a length direction corresponding to the different intervals have different pitches of the helical protrusions.

APPENDIX 9

[0121] An analysis apparatus comprising: [0122] the transport mechanism according to any one of Appendices 1 to 8; and [0123] a plurality of processing units disposed to correspond to the plurality of stop positions and performing predetermined processing on a specimen accommodated in the analysis chip.

[0124] The disclosure of Japanese Patent Application No. 2022-192253 filed on Nov. 30, 2022 is incorporated in the present specification by reference. All documents, patent applications, and

technical standards mentioned in the present specification are incorporated herein by reference to the same extent as in a case in which each document, each patent application, and each technical standard are specifically and individually described by being incorporated by reference.
EXPLANATION OF REFERENCES

Claims

1. A transport mechanism that transports an analysis chip including a body part and a pair of engagement ribs provided to protrude outward from both ends in a width direction of the body part in a transport direction orthogonal to the width direction, the transport mechanism comprising: a pair of screw shafts disposed parallel to each other along the transport direction, each of which has a shaft body and a helical protrusion helically formed on an outer peripheral surface of the shaft body, and the pair of screw shafts propels the analysis chip in the transport direction by rotating about their respective axes while the helical protrusions are engaged with the engagement ribs; a transport table that extends along the transport direction and has a plurality of stop positions where the analysis chip is stopped; and a processor that controls rotation of the screw shafts to control movement of the analysis chip between the plurality of stop positions and stopping of the analysis chip at the plurality of stop positions.
2. The transport mechanism according to claim 1, further comprising: a positioning mechanism that positions the analysis chip, which is stopped at the stop position, at a preset position in the stop position, wherein the positioning mechanism includes a positioning pin that is inserted through a through-hole penetrating the body part of the analysis chip in an up-down direction, a positioning hole that is provided at the stop position of the transport table and into which a tip of the positioning pin inserted through the through-hole is inserted, and a lifting mechanism that lifts and lowers the positioning pin between an insertion position where the positioning pin is inserted through the through-hole and the tip is inserted into the positioning hole, and a retreat position where the positioning pin is moved upward from the insertion position and is retreated from the insertion position.
3. The transport mechanism according to claim 2, wherein the processor is configured to rotate the screw shafts in a forward direction to transport the analysis chip to the stop position and stop the analysis chip, and then reversely rotate the screw shafts to return a phase position of the helical protrusion engaged with the analysis chip to an upstream side in the transport direction within a pitch range of the helical protrusion and release the engagement between the engagement ribs of the analysis chip and the helical protrusions.
4. The transport mechanism according to claim 2, wherein, in a case of rotating the screw shaft in a forward direction to transport the analysis chip to the stop position, the processor is configured to stop the transport of the analysis chip at a position where the through-hole of the analysis chip and the positioning hole at least partially overlap each other, and a center of the through-hole is located upstream of a center of the positioning hole in the transport direction, and cause the positioning mechanism to lower the positioning pin to the insertion position to send the analysis chip to a downstream side in the transport direction within a pitch range of the helical protrusion and position the analysis chip at the preset position.
5. The transport mechanism according to claim 1, wherein a bottom surface of the analysis chip is flat, and the transport table has a transport surface on which the analysis chip is placed with the bottom surface in contact with the transport surface, and the analysis chip is transported while sliding on the transport surface.
6. The transport mechanism according to claim 1, further comprising: a pressing part that presses the engagement rib of the analysis chip placed on the transport table from above.
7. The transport mechanism according to claim 6, wherein the pressing part is provided at least at the stop position.

8. The transport mechanism according to claim 1, wherein the transport table has three or more stop positions as the plurality of stop positions, and at least one of intervals between the stop positions adjacent to each other is different from the other interval, and portions of the screw shaft in a length direction corresponding to the different intervals have different pitches of the helical protrusions.

9. An analysis apparatus comprising: the transport mechanism according to claim 1; and a plurality of processing units disposed to correspond to the plurality of stop positions and performing predetermined processing on a specimen accommodated in the analysis chip.